

MapReduce Weather Data Practical Guide (Ubuntu + Python + Hadoop)

This guide explains how to run the MapReduce Weather Data practical in Ubuntu using Python and Hadoop Streaming. The practical finds the hottest and coolest year from weather data.

Step 1: Open Ubuntu Terminal

Open the Ubuntu terminal using:
Ctrl + Alt + T

Step 2: Create Practical Folder

Run the following commands:
mkdir weather_practical
cd weather_practical

Step 3: Create Weather Data File

Create weather.txt file:
nano weather.txt

Step 4: Add Data Into weather.txt

Paste the following data:

```
2001,30
2001,40
2002,50
2002,30
2003,20
2003,25
```

Save file:
CTRL + O
Press ENTER
CTRL + X

Step 5: Create Mapper Program

Create mapper.py file:
nano mapper.py

Step 6: Paste Mapper Code

Paste the following code:

```
import sys

for line in sys.stdin:
    line = line.strip()

    year, temp = line.split(",")

    print(year + "\t" + temp)
```

Save using:
CTRL + O
ENTER
CTRL + X

Step 7: Create Reducer Program

Create reducer.py:
nano reducer.py

Step 8: Paste Reducer Code

Paste the following code:

```
import sys

current_year = None
temp_list = []

hottest_year = ""
coolest_year = ""

highest_avg = -9999
lowest_avg = 9999

for line in sys.stdin:
    line = line.strip()

    year, temp = line.split("\t")

    temp = int(temp)

    if current_year == None:
        current_year = year

    if year == current_year:
        temp_list.append(temp)

    else:
        avg = sum(temp_list) / len(temp_list)

        print(current_year, "Average Temperature =", avg)

        if avg > highest_avg:
            highest_avg = avg
            hottest_year = current_year

        if avg < lowest_avg:
            lowest_avg = avg
            coolest_year = current_year

        current_year = year
        temp_list = [temp]

if current_year != None:
    avg = sum(temp_list) / len(temp_list)
```

```
print(current_year, "Average Temperature =", avg)

if avg > highest_avg:
    highest_avg = avg
    hottest_year = current_year

if avg < lowest_avg:
    lowest_avg = avg
    coolest_year = current_year

print("\nHottest Year =", hottest_year)
print("Coolest Year =", coolest_year)

Save:
CTRL + O
ENTER
CTRL + X
```

Step 9: Give Execution Permission

Run:

```
chmod +x mapper.py
chmod +x reducer.py
```

Step 10: Run Without Hadoop (Local Testing)

Run:

```
python3 mapper.py < weather.txt | sort | python3 reducer.py
```

Step 11: Expected Output

```
2001 Average Temperature = 35.0
2002 Average Temperature = 40.0
2003 Average Temperature = 22.5

Hottest Year = 2002
Coolest Year = 2003
```

Step 12: Check Hadoop Installation

Run:

```
hadoop version
```

Step 13: Start Hadoop Services

Run the following commands:

```
start-dfs.sh
start-yarn.sh
```

Step 14: Check Hadoop Processes

Run:

```
jps
```

You should see:

```
NameNode  
DataNode  
ResourceManager  
NodeManager
```

Step 15: Create Hadoop Input Directory

Run:

```
hdfs dfs -mkdir /input
```

Step 16: Upload Weather File to HDFS

Run:

```
hdfs dfs -put weather.txt /input
```

Step 17: Run Hadoop Streaming Program

Run:

```
hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar \  
-file mapper.py \  
-mapper mapper.py \  
-file reducer.py \  
-reducer reducer.py \  
-input /input/weather.txt \  
-output /output
```

Step 18: View Output

Run:

```
hdfs dfs -cat /output/part-00000
```

Step 19: Delete Output Folder Before Running Again

Hadoop does not overwrite output folders.

If running again:

```
hdfs dfs -rm -r /output
```

Step 20: Important Viva Questions

1. What does Mapper do?
Mapper extracts year and temperature.
 2. What does Reducer do?
Reducer calculates average temperature year-wise.
 3. Why use MapReduce?
To process very large weather datasets using distributed systems.
 4. Why use sort in local execution?
Reducer requires grouped keys together.
 5. What is the key?
Year
 6. What is the value?
Temperature
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